ASIC – Based Lung Sound Separation: Performance Analysis of Adaptive Line Enhancer with Least Mean Square Algorithm Across Scaled CMOS Technologies

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***Abstract*—This paper presents the design and ASIC implementation of an Adaptive Line Enhancer (ALE) using the Least Mean Square (LMS) algorithm for separating lung and heart sounds. The architecture is developed in Verilog HDL and evaluated using three CMOS technology nodes: 180nm, 90nm, and 45nm. The proposed ALE-LMS design reports power consumption of 15.79 mW, 9.31 mW, and 2.26 mW for 180nm, 90nm, and 45nm, respectively. Gate utilization stands at 9131, 8927, and 8185 gates, while the occupied area measures 247.93 mm², 78.14 mm², and 29.54 mm² across the respective technologies. The results highlight significant improvements in power and area efficiency as the technology scales down, while consistently maintaining functional accuracy. This study confirms the feasibility of integrating the ALE-LMS system into compact, low-power ASICs, making it suitable for real-time biomedical signal processing applications.**

Keywords—Application Specific Integrated Circuit, Least Mean Square Algorithm, Adaptive Line Enhancer, Complementary Metal Oxide Semiconductor.

I. INTRODUCTION

Auscultation remains one of the most fundamental, non-invasive, and widely practiced diagnostic techniques in clinical respiratory assessment, especially in resource-constrained environments. It is routinely used to detect and evaluate pulmonary conditions such as asthma, pneumonia, bronchitis, and chronic obstructive pulmonary disease [1], [2]. However, the diagnostic accuracy of auscultation is often compromised by overlapping heart sound signals (HSS) and environmental or background noise [3]. To overcome this challenge, researchers have explored various signal processing strategies aimed at enhancing lung sound signals (LSS) and suppressing unwanted interference.

Conventional digital filtering methods, such as bandpass and notch filters, often rely on fixed coefficients and perform sub-optimally in non-stationary or dynamic signal conditions. Consequently, adaptive filtering algorithms have gained popularity for their ability to adjust in real time according to the nature of incoming signals [4,5]. Among these, the Least Mean Square (LMS) algorithm is especially notable due to its simplicity, low computational requirements, and robustness across a variety of biomedical signal enhancement applications [6,11].

An efficient structure that utilizes the LMS algorithm is the Adaptive Line Enhancer (ALE), which eliminates the need for an external reference signal. ALE uses signal prediction techniques to filter out periodic and non-stationary noise components, thereby improving the clarity of lung sounds [7,10]. While software-based implementations of such algorithms have shown promising results, hardware realization is crucial for real-time applications—especially in wearable or portable medical devices where latency and power consumption are critical constraints [6,8].

Field-Programmable Gate Arrays (FPGAs) and Application-Specific Integrated Circuits (ASICs) provide a practical solution for embedding real-time adaptive filters. However, FPGAs are often limited by power and area constraints, making ASICs more suitable for dedicated low-power biomedical applications [12]. Realizing adaptive filters like LMS on ASIC platforms demands careful consideration of hardware resources, timing closure, and physical design trade-offs. The implementation of such systems using standard digital design flows—RTL to GDSII—has been demonstrated using both open-source and commercial Electronic Design Automation (EDA) tools [14].

In this work, we present an ASIC implementation of an Adaptive Line Enhancer using the LMS algorithm for lung sound enhancement. The architecture is developed in Verilog

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HDL and synthesized using industry-standard Cadence EDA tools. The design is implemented and evaluated across three technology nodes—180 nm, 90 nm, and 45 nm—to study the effect of process scaling on area, power, and performance [8,9,13]. The simulation and layout flow include synthesis, placement, routing, and generation of the GDSII file for chip fabrication.

II. DESIGN METHODOLOGY

A. *Least Mean Square Algorithm (LMS)*

LMS algorithm is one of the most widely adopted adaptive filtering techniques in DSP. It is particularly effective in scenarios where the system dynamics are unknown or time-varying, including applications such as noise cancellation, channel equalization, system identification, and biomedical signal separation.

i. *LMS Filter Model*

The LMS algorithm operates based on an FIR filter structure with adjustable coefficients. The input signal x(n)is passed through the filter to produce the output signal y(n), which is then compared with the desired signal d(n) to compute the error signal e(n). The filter adapts its coefficients over time to minimize this error as demonstrated in the Fig.1.

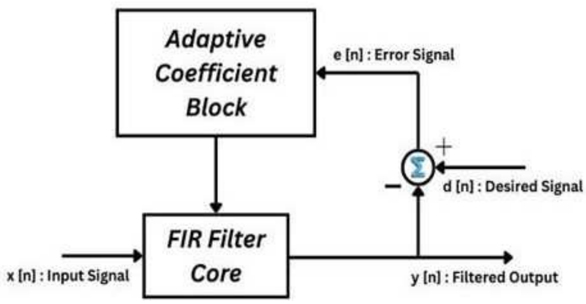

Fig.1. Structure of Least Mean Square

The output signal is defined as:

$$y(n) = \sum_{k=0}^{N-1} w_k(n) \,.\, x(n-k) \tag{1}$$

where:

N is the number of filter taps,

$w_k(n)$ represents the filter coefficient at index k and time n.

$x(n-k)$ is the delayed input sample.

The error signal is calculated as:

$$e(n) = d(n) - y(n) \tag{2}$$

ii. *Coefficient Update Rule*

The adaptive nature of the LMS filter is implemented by updating its coefficients using the error signal and input samples. The update rule follows a gradient descent approach:

$$w_k(n+1) = w_k(n) + \mu \,.\, e(n).x(n-k) \tag{3}$$

Here, μ is the step size parameter, which determines the convergence speed and stability of the filter [2,4]. A carefully chosen μ ensures that the filter converges to the optimal solution while avoiding instability.

iii. *Block Processing and State Buffer*

In block-based implementations, the LMS algorithm processes data in chunks, and the past input samples are stored in a state buffer. The required buffer length is calculated as:

$$State\ Length = N + Block\ Size - 1 \tag{4}$$

This ensures efficient filtering without repeated shifting of data. While the filter processes data in blocks, coefficient updates are still performed on a per-sample basis within each block [7].

B. *Adaptive Line Enhancer (ALE)*

The ALE is a signal processing architecture primarily designed to enhance signals corrupted by noise, especially when the noise is uncorrelated with the desired signal. The system relies on the principle that while the desired signal exhibits some degree of correlation over time, the noise does not. ALE takes advantage of this difference by utilizing a delayed version of the input signal to construct a reference signal.

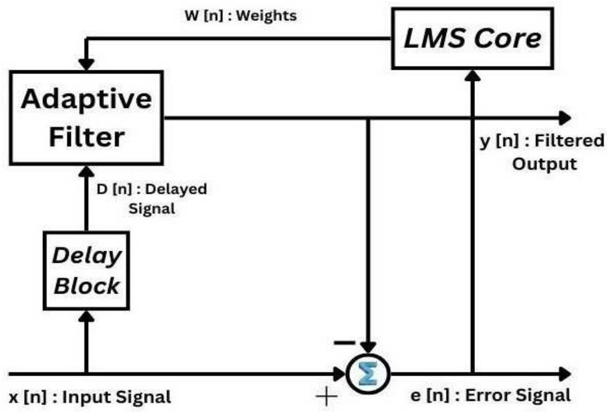

Fig.2. Adaptive Line Enhancer (ALE)

In the typical ALE architecture, as shown in Fig.2., the input signal x(n)is passed through a delay block, generating a delayed version D[n]. This delayed signal serves as an input to the adaptive filter, which attempts to predict the current signal. The output of the adaptive filter y(n)is then compared with the original signal x(n) to compute an error signal e(n), which reflects the difference between the predicted and actual signal. This error is used to iteratively adjust the adaptive filter coefficients via an adaptive algorithm, such as the LMS algorithm.

The goal of the ALE is to minimize the error signal by adapting the filter coefficients in a way that the output of the adaptive filter converges towards the noise-free component of the input. Over time, the filter effectively learns the characteristics of the signal, enabling it to suppress noise components and enhance the desired signal. The ALE is particularly useful in applications such as biomedical signal enhancement (e.g., lung or heart sounds), speech processing, and communication systems, where non-stationary noise environments are prevalent. It is computationally efficient and can be implemented in both hardware (e.g., FPGA or ASIC) and software platforms.

C. *Adaptive Line Enhnacer with Least Mean Square Algorithm (ALE – LMS)*

The proposed architecture of the ALE using the LMS algorithm is illustrated in Fig.3. The system is designed for the real-time separation of lung sounds from composite biomedical signals containing both lung and heart sounds. The ALE does not require a separate reference noise signal, making it ideal for non-invasive auscultation-based applications.

As shown in Fig.3, the system takes two primary 16-bit input signals: $Reff_{in}[15:0]$ and $Data_{in}[15:0]$. The signal $Reff_{in}$ acts as the delayed version of the input signal used for prediction, while $Data_{in}$ contains the original mixed signal, which includes lung sounds, heart sounds, and potential background noise. The signal processing starts with a Delay Register block, which holds the Data in for alignment and supplies it to both the Adaptive FIR Filter and the Adaptive FIR Filter Coefficients block. The Adaptive FIR Filter predicts the signal by processing the delayed version of the input, and its coefficients are dynamically updated through the LMS algorithm.

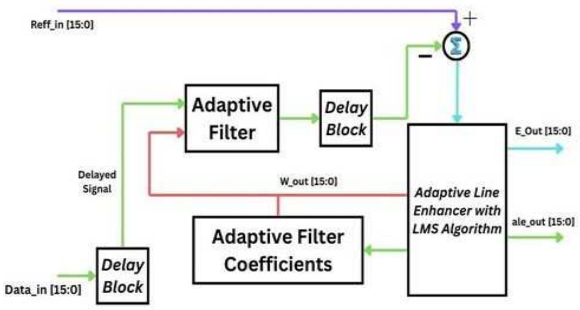

Fig.3. Implementation of ALE-LMS in ASIC

The filter output is passed through another Delay Reg block to maintain synchronization and timing accuracy. This output is then fed into the Error Computation Unit, where it is subtracted from the reference signal $Reff_{in}$ [15:0] to generate the error signal E_{out} [15:0] . This error signal is a key parameter for adaptive coefficient updates, which are handled by the Adaptive Line Enhancer with LMS Algorithm block.

The ALE with LMS Algorithm block utilizes both the error signal and the predicted output to iteratively adjust the coefficients for optimal filtering. The final enhanced lung sound signal is output through $ALE_{out}[15:0]$, while the instantaneous error output is provided as $E_{out}[15:0]$. This architecture is designed using Verilog HDL and further synthesized using Cadence tools. The complete design is implemented as an ASIC in multiple CMOS technology nodes (180 nm, 90 nm, and 45 nm) to evaluate its performance in terms of power consumption, area efficiency, and speed. The GDSII layout generated through the Cadence physical design flow validates the silicon realization of the proposed system.

III. Implementation

A. *Verilog HDL*

The hardware realization of the ALE using the LMS algorithm was carried out through Verilog Hardware Description Language (HDL). Verilog was selected due to its robustness and widespread use in digital design, particularly for ASIC and FPGA development. The complete ALE system was modularized into various functional blocks such as the delay unit, FIR filter, error computation unit, and coefficient update module. Each block was designed using structured style. The implementation employed 16-bit fixed-point representation ([15:0]) to ensure efficient hardware utilization while maintaining signal precision, which is crucial in biomedical applications. The LMS algorithm was realized using a sample-by-sample adaptation mechanism. The filter coefficients were dynamically updated based on the instantaneous error signal, and the step-size (μ) was set as a configurable parameter to control the convergence speed.

Comprehensive simulation and verification were performed using HDL simulation tools. Dedicated testbenches were developed to validate the functionality of each module with representative input patterns such as synthetic lung and heart sounds. Following functional verification, the RTL code was synthesized for ASIC implementation. The synthesized netlist was further integrated into a complete digital flow using Cadence tools. This included stages like floor planning, power planning, placement, clock tree synthesis, routing, and generation of the final GDSII file. This implementation demonstrates a complete RTL-to-GDSII flow, enabling the realization of the ALE-LMS algorithm in silicon for efficient real-time biomedical signal processing.

B. *ASIC Design Flow*

The RTL to GDSII design flow comprises several key stages, each of which plays a crucial role in the physical realization of the proposed architecture. The following subsections describe the main steps followed during the ASIC implementation of the ALE-LMS design using Cadence tools.

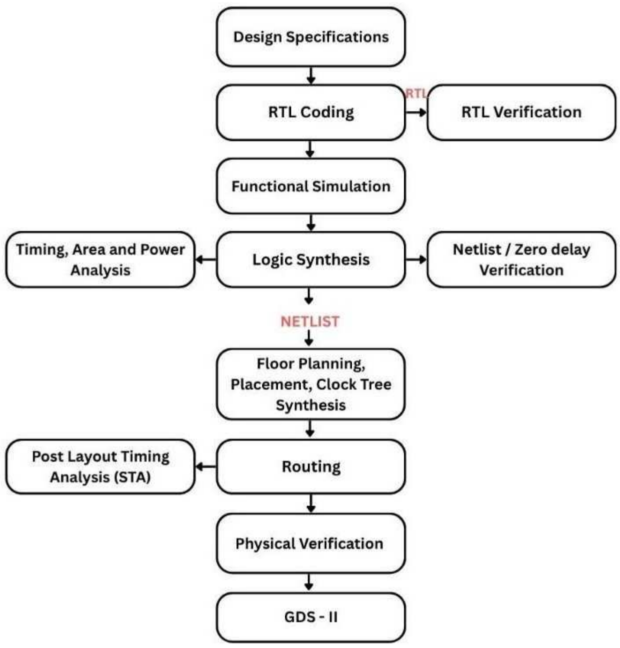

Fig 4: ASIC Design Flow

The ASIC design flow follows a structured sequence that transforms initial specifications into a fabricated silicon chip, as illustrated in Fig. 4.The process begins with defining the design specifications, followed by RTL coding using Verilog or VHDL. RTL verification ensures that the functionality matches the intended behaviour. After verification, functional simulation is performed to validate logical correctness without considering timing. The verified RTL code is synthesized into a gate-level netlist through logic synthesis. Post-synthesis, timing, area, and power analyses are conducted, along with netlist/zero delay verification. The netlist then advances to the physical design phase, starting with floor planning, placement, and clock tree synthesis (CTS) to organize cells and distribute the clock signal effectively. Subsequent routing establishes physical connections, followed by static timing analysis (STA) to ensure timing requirements are met. After timing closure, physical verification involving Design Rule Check (DRC) and Layout Versus Schematic (LVS) confirms manufacturability. The finalized design is exported in GDS-II format for fabrication. This systematic flow ensures that the ASIC design is functionally correct, meets all constraints, and is ready for tape-out.

C. Technology Comparison

In this work, the ALE using the LMS algorithm has been implemented in three different CMOS technology nodes: 180 nm, 90 nm, and 45 nm. Each technology offers different trade-offs in terms of power consumption, speed, area efficiency, and leakage current. A comparative analysis of these nodes is crucial to identify the most suitable process for optimized ASIC implementation. Table I summarizes the key characteristics of the selected technologies and highlights how process scaling impacts overall design performance.

Table I. Comparison of different CMOS technology nodes (180nm, 90nm, 45nm)

Parameter	**180 nm**	**90 nm**	**45 nm**
Operating Voltage	~ 1.8 V – 3.3 V	~ 1.0 V – 1.2 V	~0.9 V – 1.1 V
Gate Delay	~20 – 40 ps	~10 – 20 ps	~5 – 10 ps
Power Consumption	High	Medium	Low
Leakage Current	Very Low	Moderate	High
Die Area (Same for Logic)	Large	Medium	Small
Speed (Performance)	Moderate	High	Very High
Cost per mm^2 (approx.)	Low	Medium	High
Integration Density	~0.5M gates/mm^2	~1.5M gates/mm^2	~3M gates/mm^2

IV. Results And Discussion

This section analyzes the ALE-LMS design, implemented in Verilog and synthesized using 180 nm, 90 nm, and 45 nm CMOS to compare area, power, and timing. Fig.5 shows a simulation waveform from Cadence, verifying key signals like clock, reset, input, and output after GDSII generation.

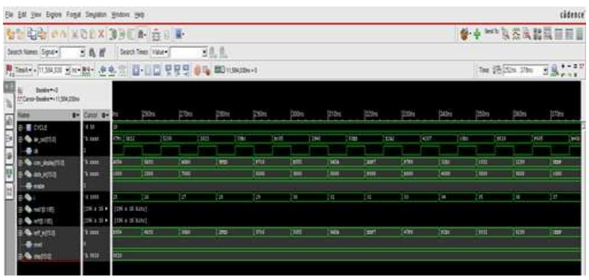

Fig.5. Functional Simulation Waveform of ALE–LMS Architecture

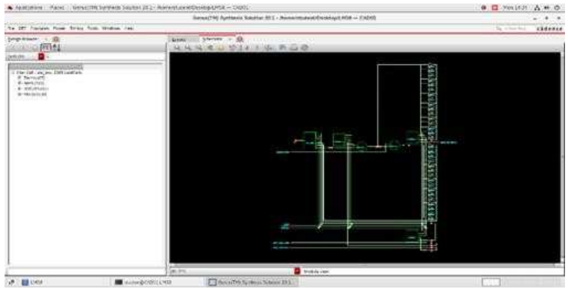

Fig.6. Schematic View of ALE-LMS Architecture

The waveforms confirm correct data flow and logic operation across clock cycles during RTL simulation. Fig.6 shows the synthesized ALE-LMS schematic in Cadence Genus, illustrating the gate-level netlist with interconnected logic blocks like multiplexers, adders, and registers. The left panel shows the Design Browser for navigating the module hierarchy, while the right displays the schematic with signal paths and I/O pins. This view helps verify logic, analyze hierarchy, and understand data flow before physical layout

A. 45nm Technology

An ALE using the LMS algorithm was implemented in 45 nm CMOS to assess efficiency in power, timing, and area. The design, verified with Cadence tools, shows strong potential for low-power biomedical applications. The Fig.7 45nm ALE-LMS layout shows optimized placement, routing, and power grids. Final checks like DRC, LVS, and timing ensure readiness for GDSII and fabrication.

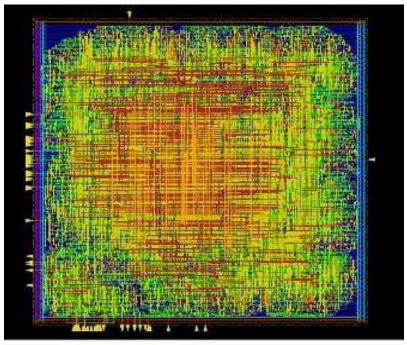

Fig.7. Final Design of ALE – LMS in 45nm technology

B. 90nm Technology

The Fig.8 ALE-LMS filter was implemented using 90 nm CMOS to assess area, power, and gate usage, serving as a baseline before moving to smaller nodes. Results from synthesis and layout offer insights into power, size, and efficiency, showcasing how the design performs on a mature, well-supported process.

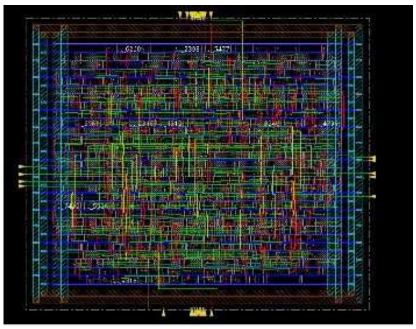

Fig.8. Final Design of ALE-LMS in 90nm Technology

C. *180nm Technology*

The Fig.9 ALE-LMS filter was implemented in 180 nm CMOS to assess its performance for cost-sensitive, low-power applications. This analysis highlights the design's behavior with larger feature sizes, offering insights into power, area, and efficiency for legacy or resource-constrained systems.

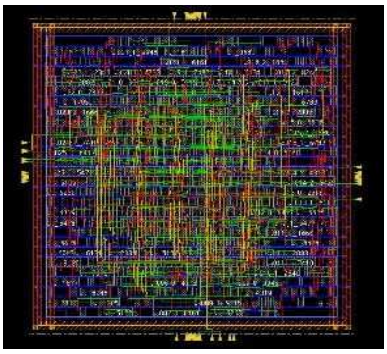

Fig.9. Final Design of ALE-LMS in 180nm Technology

D. *Analysis of Results*

The Table II shows the variation in leakage, dynamic, and total power across 180 nm, 90 nm, and 45 nm nodes. As technology scales, dynamic power decreases, while leakage power varies based on transistor characteristics, affecting overall energy efficiency.

Table II. Power Consumption Analysis of ALE-LMS 8 across Multiple CMOS Technology Nodes

Technology Node	Leakage Power (mW)	Dynamic Power (mW)	Total Power (mW)
180 nm	0.7675	15.03	15.7976
90 nm	0.3425	8.9768	9.3193
45 nm	0.001	2.2655	2.2665

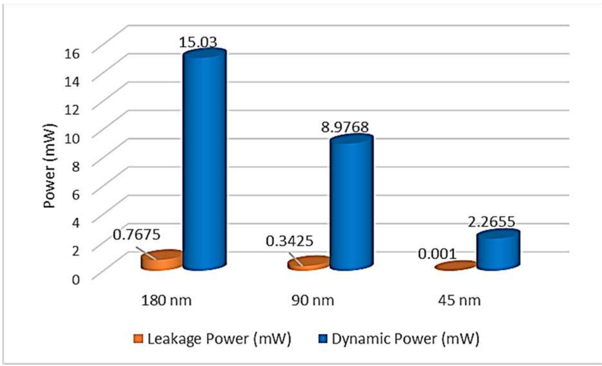

Fig.10. Power Analysis Across scaled CMOS Technologies

This Fig.10 compares the total power consumption of the ALE-LMS architecture across 180nm, 90nm, and 45nm technologies. Both dynamic and leakage power decrease significantly with scaling, confirming the energy efficiency advantages of smaller nodes. The Table III compares area usage of sequential elements, inverters, and logic gates across 180 nm, 90 nm, and 45 nm nodes, showing reduced area with scaling. This highlights how smaller nodes enable higher integration in chip design.

Table III. Area Metrics of ALE-LMS 8 Design

Parameter	Area (mm^2)		
	180 nm	**90 nm**	**45 nm**
Sequential	23.950	7.193	2.913
Inverter	5.415	1.230	0.297
Logic	218.571	69.724	26.338

This Fig.11 shows the chip area consumed by the design in each technology. The results clearly indicate that smaller process nodes enable more compact layouts, which is beneficial for integration in size-constrained systems. The Table IV shows gate counts for the 8-tap LMS filter across 180 nm, 90 nm, and 45 nm nodes. While sequential gates remain constant, variations in inverter and logic gates reflect differences in synthesis and optimization during scaling.

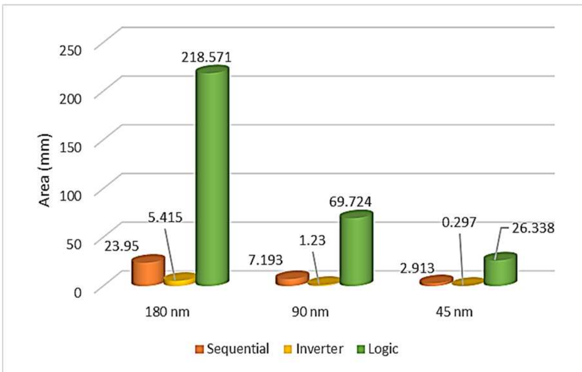

Fig.11. Area Metrics Across scaled CMOS Technologies

Table.IV. Gate Count analysis of ALE-LMS 8 Design

Parameter	Gates		
	180 nm	**90 nm**	**45 nm**
Sequential	288	288	288
Inverter	812	542	407
Logic	7899	8097	7490

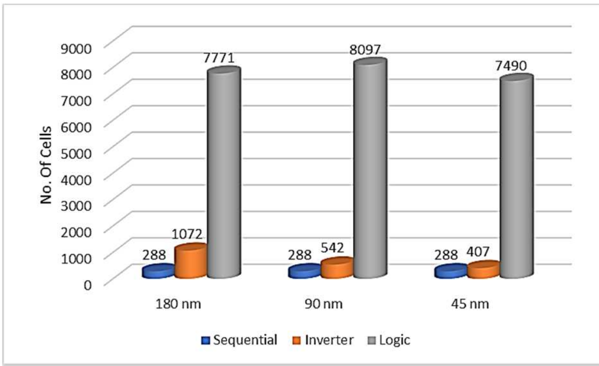

Fig.12. Gate Count Analysis Across scaled CMOS Technologies

This Fig.12 shows the chip area consumed by the design in each technology. The results clearly indicate that smaller process nodes enable more compact layouts, which is beneficial for integration in size-constrained systems.

Table V. Overall Comparative analysis of Power, Area and Gate count Across CMOS Technologies

Parameter	180nm	90nm	45nm
Power (mW)	15.7976	9.3193	2.2665
Gates	9131	8927	8185
Area (mm^2)	247.936	78.147	29.548

The Table V. compares power, gate, and area utilization across 180 nm, 90 nm, and 45 nm nodes, showing reduced power and improved efficiency with technology scaling. The Fig.13 shows power consumption is highest in 180nm and decreases significantly at 90nm and 45nm, confirming improved energy efficiency with scaling. The Fig.14 shows gate utilization shows minimal variation across nodes, with a slight reduction in 45nm, indicating consistent synthesis and efficient logic mapping.

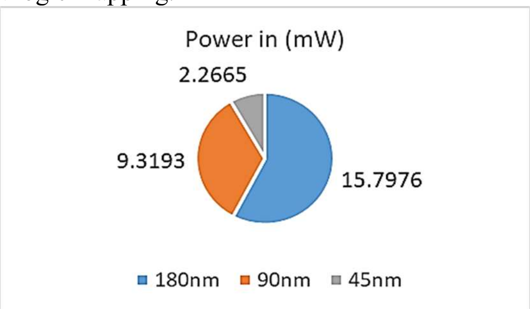

Fig.13. Power Consumption of ALE-LMS across different CMOS

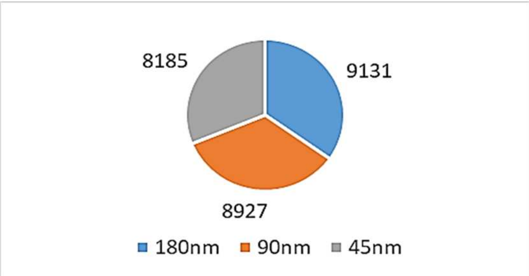

Fig.14. Gate Utilization of ALE-LMS across different CMOS Technologies

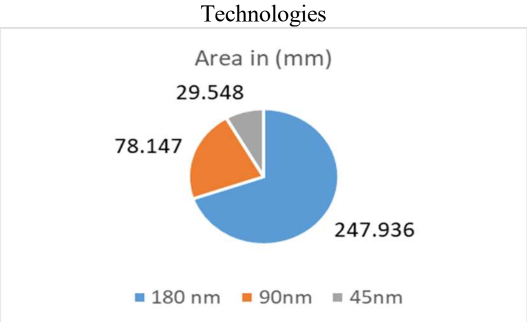

Fig.15. Area Utilization of ALE-LMS across different CMOS Technologies

The Fig.15 shows area usage decreases clearly from 180nm to 45nm, highlighting the advantage of advanced nodes for compact, silicon-efficient ASIC designs.

V. Conclusion

This paper presents the design and ASIC implementation of an Adaptive Line Enhancer (ALE) based on the Least Mean Square (LMS) algorithm for the separation of lung and heart sounds. The architecture was synthesized using 180nm, 90nm, and 45nm CMOS technology nodes. The design achieves power consumptions of 15.79 mW, 9.31 mW, and 2.26 mW for 180nm, 90nm, and 45nm, respectively. The occupied chip areas are measured as 247.93 mm², 78.14 mm², and 29.54 mm², while the gate utilizations are reported as 9131, 8927, and 8185 gates across the respective nodes. Among the evaluated technologies, the 45nm implementation demonstrates the highest efficiency in terms of reduced power consumption and minimized silicon area, without compromising functional accuracy. These results confirm the feasibility and effectiveness of integrating the ALE-LMS system into a dedicated low-power ASIC specifically for real-time lung sound separation.

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